BIOL 180 - Lecture 6: The Cell Membrane and Transport

Structure
The plasma membrane is a phospholipid bilayer. Each phospholipid has a hydrophilic head and two hydrophobic fatty acid tails, so the tails face inward and the heads face the water on both sides. Proteins embedded in the bilayer move laterally; this is the fluid mosaic model. Cholesterol keeps the membrane fluid at low temperatures and limits movement at high temperatures.

Selective permeability
Small nonpolar molecules (O2, CO2) cross the bilayer directly. Ions and large polar molecules (glucose) need transport proteins.

Passive transport (no energy required)
- Diffusion: molecules move down their concentration gradient.
- Osmosis: water diffuses across a membrane toward the side with more solute.
- Facilitated diffusion: channel or carrier proteins help molecules move down their gradient, for example aquaporins for water.

Tonicity
A cell in a hypotonic solution gains water (animal cells may lyse; plant cells become turgid). In a hypertonic solution the cell loses water (plasmolysis in plant cells). An isotonic solution causes no net movement.

Active transport (requires ATP)
The sodium-potassium pump moves 3 Na+ out and 2 K+ in per ATP, against both gradients. This builds the membrane potential that nerve cells use to send signals.

Bulk transport
Exocytosis releases materials using vesicles that fuse with the membrane. Endocytosis takes materials in: phagocytosis (cell eating), pinocytosis (cell drinking), and receptor-mediated endocytosis, which is specific to the molecules bound by receptors.